

APSMO 2010 Set 1

1 Problems

Problem 1.1 What is the value of the following?

$$(8 \times 4) + (8 \times 3) + (8 \times 2) + (8 \times 1)$$

Problem 1.2 A bag contains 18 jelly beans. 4 are red, 6 are white and 8 are blue. Amanda takes them out one at a time without looking.

What is the fewest jelly beans she must take out to be certain that at least 2 of the jelly beans she takes out are blue?

Problem 1.3 A prime number is a counting number with exactly two factors, the number itself and the number 1. In the sequence 2, 5, 11, 23, $\dots$, each number is obtained by doubling the previous number and adding 1.

What is the first number in the sequence that is *not* a prime number?

Problem 1.4 A digital timer counts down from 5 minutes (5 : 00) to 0 : 00 one second at a time.

For how many seconds does at least one of the three digits show a 2?

Problem 1.5 A rectangular box has a top that is 15 cm by 20 cm and a height of 4 cm. An ant begins at one corner of the box and walks along the edges. It touches all eight corners.

What is the shortest distance, in cm, that the ant could travel?